Numerical Analysis Homework 3

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I. Cubic Spline Determination

Let $p(x) = ax^3 + bx^2 + cx + d$. From $s(0) = p(0) = 0$, we obtain:

$$d = 0.$$

At the knot $x = 1$, we require continuity of function value, first derivative, and second derivative:

$$\begin{aligned} p(1) &= (2-1)^3 = 1, \\ p'(1) &= \frac{d}{dx}(2-x)^3\bigg|_{x=1} = -3, \\ p''(1) &= \frac{d^2}{dx^2}(2-x)^3\bigg|_{x=1} = 6. \end{aligned}$$

Compute derivatives of p:

$$p'(x) = 3ax^2 + 2bx + c, \quad p''(x) = 6ax + 2b.$$

Substitute the conditions at $x = 1$:

$$\begin{aligned} a + b + c &= 1, \\ 3a + 2b + c &= -3, \\ 6a + 2b &= 6. \end{aligned}$$

$$\implies a = 7, \quad b = -18, \quad c = 12.$$

Thus,

$$p(x) = 7x^3 - 18x^2 + 12x.$$

Since $s''(0) = p''(0) = 6a \cdot 0 + 2b = -36 \neq 0$, $s(x)$ is not a natural cubic spline.

II. Quadratic Spline Interpolation

II-a

A quadratic spline $s \in \mathbb{S}_2^1$ on $[a, b]$ with nodes $a = x_1 < x_2 < \cdots < x_n = b$ consists of $n - 1$ quadratic polynomials p_i on each interval $[x_i, x_{i+1}]$. Each p_i has 3 parameters, giving a total of $3(n-1)$ parameters. The conditions are:

- Interpolation: $p_i(x_i) = f_i$ and $p_i(x_{i+1}) = f_{i+1}$ for $i = 1, \ldots, n-1$, yielding $2(n-1)$ conditions.
- Continuity of first derivative at interior nodes: $p'_{i-1}(x_i) = p'_i(x_i)$ for $i = 2, \ldots, n-1$, yielding $n - 2$ conditions.

Total conditions: $2(n-1) + (n-2) = 3n - 4$. Since $3(n-1) = 3n - 3$, there is one degree of freedom. Hence, an additional condition is needed to determine s uniquely.

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II-b

Let $p_i(x) = A_i(x - x_i)^2 + B_i(x - x_i) + C_i$. The conditions are:

$$\begin{aligned} p_i(x_i) &= C_i = f_i, \\ p_i'(x_i) &= B_i = m_i, \\ p_i(x_{i+1}) &= A_i(\Delta x_i)^2 + m_i \Delta x_i + f_i = f_{i+1}, \quad \text{where } \Delta x_i = x_{i+1} - x_i. \end{aligned}$$

Solving for A_i:

$$A_i = \frac{f_{i+1} - f_i - m_i \Delta x_i}{(\Delta x_i)^2}.$$

Thus,

$$p_i(x) = f_i + m_i(x - x_i) + \frac{f_{i+1} - f_i - m_i \Delta x_i}{(\Delta x_i)^2}(x - x_i)^2.$$

II-c

Given $m_1 = f'(a)$, we use the continuity of the first derivative at interior nodes. For $i = 2, \dots, n-1$:

$$p_{i-1}'(x_i) = p_i'(x_i).$$

From part (b), on $[x_{i-1}, x_i]$:

$$p_{i-1}'(x) = 2A_{i-1}(x - x_{i-1}) + m_{i-1}, \quad \text{with } A_{i-1} = \frac{f_i - f_{i-1} - m_{i-1}\Delta x_{i-1}}{(\Delta x_{i-1})^2}.$$

At $x = x_i$:

$$p_{i-1}'(x_i) = 2 \cdot \frac{f_i - f_{i-1} - m_{i-1}\Delta x_{i-1}}{\Delta x_{i-1}} + m_{i-1} = \frac{2(f_i - f_{i-1})}{\Delta x_{i-1}} - m_{i-1}.$$

Since $p_i'(x_i) = m_i$, we obtain:

$$m_i = \frac{2(f_i - f_{i-1})}{\Delta x_{i-1}} - m_{i-1}, \quad \text{for } i = 2, \dots, n-1.$$

Starting from m_1, we can compute $m_2, m_3, \dots, m_{n-1}$ recursively.

III. Natural Cubic Spline Construction

Let $s_2(x) = Ax^3 + Bx^2 + Cx + D$. The conditions are:

By continuity at $x = 0$:

$$s_1(0) = s_2(0) \implies 1 + c = D, \tag{1}$$
$$s_1'(0) = s_2'(0) \implies 3c = C, \tag{2}$$
$$s_1''(0) = s_2''(0) \implies 6c = 2B \implies B = 3c. \tag{3}$$

By natural boundary condition at $x = 1$:

$$s_2''(1) = 6A + 2B = 6A + 6c = 0 \implies A = -c. \tag{4}$$

Thus,

$$s_2(x) = -cx^3 + 3cx^2 + 3cx + (1 + c).$$

Additional condition $s(1) = -1$:

$$s_2(1) = -c + 3c + 3c + 1 + c = 6c + 1 = -1 \implies c = -\frac{1}{3}.$$

IV. Natural Cubic Spline and Bending Energy

IV-a

Let

$$s(x) = \begin{cases} s_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1 & \text{for } x \in [-1, 0], \\ s_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2 & \text{for } x \in [0, 1]. \end{cases}$$

Given $f(-1) = 0$, $f(0) = 1$, $f(1) = 0$, the conditions are:

$$s_1(-1) = 0, \quad s_1(0) = 1, \quad s_2(0) = 1, \quad s_2(1) = 0, \quad s_1'(0) = s_2'(0), \quad s_1''(0) = s_2''(0), \quad s_1''(-1) = 0, \quad s_2''(1) = 0$$

Solving the equations gives:

$$a_1 = -\frac{1}{2}, \quad b_1 = -\frac{3}{2}, \quad c_1 = 0, \quad a_2 = \frac{1}{2}, \quad b_2 = -\frac{3}{2}, \quad c_2 = 0.$$

Thus,

$$s(x) = \begin{cases} 1 - \frac{1}{2}x^3 - \frac{3}{2}x^2 & \text{for } x \in [-1, 0], \\ 1 + \frac{1}{2}x^3 - \frac{3}{2}x^2 & \text{for } x \in [0, 1]. \end{cases}$$

IV-b

The bending energy is $E(g) = \int_{-1}^{1} [g''(x)]^2 dx$.

For the natural cubic spline s:

$$s_1''(x) = -3x - 3, \quad s_2''(x) = 3x - 3$$

Thus

$$E(s) = \int_{-1}^{0} (-3x - 3)^2 dx + \int_{0}^{1} (3x - 3)^2 dx = 6$$

(i) For the quadratic interpolant $q(x) = 1 - x^2$:

$$q''(x) = -2, \quad E(q) = \int_{-1}^{1} (-2)^2 dx = 4 \cdot 2 = 8.$$

(ii) For $f(x) = \cos\left(\frac{\pi}{2}x\right)$:

$$f''(x) = -\frac{\pi^2}{4} \cos\left(\frac{\pi}{2}x\right), \quad [f''(x)]^2 = \frac{\pi^4}{16} \cos^2\left(\frac{\pi}{2}x\right),$$

$$E(f) = \frac{\pi^4}{16} \int_{-1}^{1} \cos^2\left(\frac{\pi}{2}x\right) dx = \frac{\pi^4}{16} \cdot 2 \int_{0}^{1} \cos^2\left(\frac{\pi}{2}x\right) dx.$$

Using $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$,

$$\int_{0}^{1} \cos^2\left(\frac{\pi}{2}x\right) dx = \frac{1}{2} \int_{0}^{1} (1 + \cos(\pi x))\, dx = \frac{1}{2},$$

so $E(f) = \frac{\pi^4}{16} \cdot 2 \cdot \frac{1}{2} = \frac{\pi^4}{16} \approx 6.088$.

We have $E(s) = 6 < E(q) = 8$ and $E(s) = 6 < E(f) \approx 6.088$

V. The Quadratic B-spline $B_i^2(x)$

V-a

Using the recursive definition from Definition 3.23 with the base case:

$$B_i^0(x) = \begin{cases} 1 & \text{if } x \in (t_{i-1}, t_i], \\ 0 & \text{otherwise.} \end{cases}$$

For degree 1 (hat functions):

$$B_i^1(x) = \frac{x - t_{i-1}}{t_i - t_{i-1}} B_i^0(x) + \frac{t_{i+1} - x}{t_{i+1} - t_i} B_{i+1}^0(x).$$

For degree 2:

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} B_i^1(x) + \frac{t_{i+2} - x}{t_{i+2} - t_i} B_{i+1}^1(x).$$

Now compute explicitly on each interval:

Interval 1: $x \in (t_{i-1}, t_i]$
Here $B_i^1(x) = \frac{x - t_{i-1}}{t_i - t_{i-1}}$ and $B_{i+1}^1(x) = 0$. So:

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{x - t_{i-1}}{t_i - t_{i-1}} = \frac{(x - t_{i-1})^2}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}.$$

Interval 2: $x \in (t_i, t_{i+1}]$
Here:

$$B_i^1(x) = \frac{t_{i+1} - x}{t_{i+1} - t_i}, \quad B_{i+1}^1(x) = \frac{x - t_i}{t_{i+1} - t_i}.$$

So:

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} \cdot \frac{t_{i+1} - x}{t_{i+1} - t_i} + \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{x - t_i}{t_{i+1} - t_i}.$$

Interval 3: $x \in (t_{i+1}, t_{i+2}]$
Here $B_i^1(x) = 0$ and $B_{i+1}^1(x) = \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}}$. So:

$$B_i^2(x) = \frac{t_{i+2} - x}{t_{i+2} - t_i} \cdot \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}} = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}.$$

Interval 4: $x \le t_{i-1}$ **or** $x \ge t_{i+1}$
Obviously, $B_i^2(x) = 0$

This matches the expression in Example 3.25.

V-b

By calculating the derivative of $B_i^2(x)$:

$$\frac{d}{dx} B_i^2(x) = \begin{cases} \frac{2(x - t_{i-1})}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{t_{i+1} + t_{i-1} - 2x}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{2x - t_i - t_{i+2}}{(t_{i+2} - t_i)(t_{i+1} - t_i)}, & x \in (t_i, t_{i+1}] \\ -\frac{2(t_{i+2} - x)}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0, & \text{otherwise} \end{cases}$$

It is easy to verify that

$$\frac{d}{dx} B_i^2(t_i^-) = \frac{2}{t_{i+1} - t_{i-1}} = \frac{d}{dx} B_i^2(t_i^+)$$

$$\frac{d}{dx} B_i^2(t_{i+1}^-) = -\frac{2}{t_{i+2} - t_i} = \frac{d}{dx} B_i^2(t_{i+1}^+)$$

V-c

According to the expression in V-b, it is obvious that $\frac{d}{dx} B_i^2(x) \neq 0$ for $x \in (t_{i-1}, t_i]$ and $x \in (t_{i+1}, t_{i+2}]$

And when $x \in (t_i, t_{i+1}]$, $\frac{d}{dx} B_i^2(x) = 0$ yields

$$\frac{t_{i+1} + t_{i-1} - 2x}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} + \frac{2x - t_i - t_{i+2}}{(t_{i+2} - t_i)(t_{i+1} - t_i)} = 0$$

$$\implies x^* = \frac{t_i t_{i+1} - t_{i-1} t_{i+2}}{(t_i + t_{i+1}) - (t_{i-1} + t_{i+2})} \in (t_i, t_{i+1}]$$

V-d

According to the analysis above, $\frac{d}{dx} B_i^2(x) > 0, x \in (t_{i-1}, x^*)$ and $\frac{d}{dx} B_i^2(x) < 0, x \in (x^*, t_{i+2})$

Thus $B_i^2(x)$ increases in (t_{i-1}, x^*) and decreases in (x^*, t_{i+2})

Since $B_i^2(t_{i-1}) = 0 = B_i^2(t_{i+2})$, we obtain that $B_i^2(x) > 0, x \in (t_{i-1}, t_{i+2}), i \in \mathbb{Z}$

By Corollary 3.48, $\sum_i B_i^2(x) = 1$, so $B_i^2(x) < 1$

V-e

For $t_i = i$, the quadratic B-spline is:

$$B_i^2(x) = \begin{cases} \frac{(x-(i-1))^2}{2} & x \in [i-1, i], \\ -(x-i)^2 + (x-i) + \frac{1}{2} & x \in [i, i+1], \\ \frac{(i+2-x)^2}{2} & x \in [i+1, i+2], \\ 0 & \text{otherwise.} \end{cases}$$

VI. Theorem 3.32

$$\begin{aligned}
&(t_{i+2} - t_{i-1})[t_{i-1}, t_i, t_{i+1}, t_{i+2}](t-x)_+^2 \\
=&(t_{i+2} - t_{i-1})\frac{[t_i, t_{i+1}, t_{i+2}] - [t_{i-1}, t_i, t_{i+1}]}{t_{i+2} - t_{i-1}}(t-x)_+^2 \\
=&[t_i, t_{i+1}, t_{i+2}](t-x)_+^2 - [t_{i-1}, t_i, t_{i+1}](t-x)_+^2 \\
=&\frac{[t_{i+1}, t_{i+2}](t-x)_+^2 - [t_i, t_{i+2}](t-x)_+^2 - [t_{i-1}, t_{i+1}](t-x)_+^2 + [t_{i-1}, t_i](t-x)_+^2}{t_{i+1} - t_i} \\
=&\frac{1}{t_{i+1} - t_i}\left(\frac{(t_{i+2} - x)_+^2 - (t_{i+1} - x)_+^2}{t_{i+2} - t_{i+1}} - \frac{(t_{i+2} - x)_+^2 - (t_i - x)_+^2}{t_{i+2} - t_i} - \frac{(t_{i+1} - x)_+^2 - (t_{i-1} - x)_+^2}{t_{i+1} - t_{i-1}} + \frac{(t_{i-1} - x)_+^2 - (t_i - x)_+^2}{t_{i-1} - t_i}\right) \\
=&\begin{cases} \frac{(x-t_{i-1})^2}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}, & x \in (t_{i-1}, t_i] \\ \frac{x-t_{i-1}}{t_{i+1}-t_{i-1}} \cdot \frac{t_{i+1}-x}{t_{i+1}-t_i} + \frac{t_{i+2}-x}{t_{i+2}-t_i} \cdot \frac{x-t_i}{t_{i+1}-t_i}, & x \in (t_i, t_{i+1}] \\ \frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}, & x \in (t_{i+1}, t_{i+2}] \\ 0, & \text{otherwise} \end{cases} \\
=&B_i^2(x)
\end{aligned}$$

VII. Scaled Integral of B-splines

Define $K_i^n(x) = \dfrac{1}{t_{i+n} - t_{i-1}} \int_{t_{i-1}}^{t_{i+n}} B_i^n(x)\mathrm{d}x$

We use the theorem on derivatives of B-splines

$$\frac{\mathrm{d}}{\mathrm{d}x}B_i^n(x) = \frac{nB_i^{n-1}(x)}{t_{i+n-1} - t_{i-1}} - \frac{nB_{i+1}^{n-1}(x)}{t_{i+n} - t_i}$$

Integrating both sides over $\mathbb{R}$, we obtain:

$$\int_{t_{i-1}}^{t_{i+n}} \frac{\mathrm{d}}{\mathrm{d}x}B_i^n(x)\mathrm{d}x = \int_{t_{i-1}}^{t_{i+n-1}} \frac{nB_i^{n-1}(x)}{t_{i+n-1} - t_{i-1}}\mathrm{d}x - \int_{t_i}^{t_{i+n}} \frac{nB_{i+1}^{n-1}(x)}{t_{i+n} - t_i}$$

Since $B_i^n(x)$ is compactly supported, the left-hand side is zero. Thus,

$$0 = \frac{n}{t_{i+n} - t_i}\int_{-\infty}^{\infty} B_i^{n-1}(x)dx - \frac{n}{t_{i+n+1} - t_{i+1}}\int_{-\infty}^{\infty} B_{i+1}^{n-1}(x)dx,$$

which implies

$$K_i^{n-1}(x) = K_{i+1}^{n-1}(x)$$

So $K_i^n(x)$, which is the scaled integral of a B-spline $B_i^n(x)$ over its support, is independent of the index i.

VIII. Symmetric Polynomials

VIII-a

The theorem states that for any positive integer m, any integer i, and any $n = 0, 1, \dots, m$, the complete symmetric polynomial can be expressed as a divided difference:

$$\tau_{m-n}(x_i, \dots, x_{i+n}) = [x_i, \dots, x_{i+n}]x^m.$$

We verify this for $m = 4$ and $n = 2$. Then $m - n = 2$, and we consider variables x_i, x_{i+1}, x_{i+2}. The complete symmetric polynomial of degree 2 in three variables is:

$$\tau_2(x_i, x_{i+1}, x_{i+2}) = x_i^2 + x_i x_{i+1} + x_i x_{i+2} + x_{i+1}^2 + x_{i+1}x_{i+2} + x_{i+2}^2.$$

Now compute the divided difference $[x_i, x_{i+1}, x_{i+2}]x^4$. Let $a = x_i$, $b = x_{i+1}$, $c = x_{i+2}$ for simplicity. The divided difference is defined recursively:

$$[a, b, c]x^4 = \frac{[b, c]x^4 - [a, b]x^4}{c - a}.$$

First, compute the first-order divided differences:

$$[a, b]x^4 = \frac{b^4 - a^4}{b - a} = a^3 + a^2 b + ab^2 + b^3,$$

$$[b, c]x^4 = \frac{c^4 - b^4}{c - b} = b^3 + b^2 c + bc^2 + c^3.$$

Then,

$$[a, b, c]x^4 = \frac{(b^3 + b^2 c + bc^2 + c^3) - (a^3 + a^2 b + ab^2 + b^3)}{c - a} = \frac{b^2 c + bc^2 + c^3 - a^3 - a^2 b - ab^2}{c - a}.$$

Let $M = b^2 c + bc^2 + c^3 - a^3 - a^2 b - ab^2$. Since M vanishes when $a = c$, $c - a$ is a factor. Polynomial division of M by $c - a$ yields:

$$\frac{M}{c - a} = a^2 + ab + ac + b^2 + bc + c^2.$$

Thus,

$$[a, b, c]x^4 = a^2 + ab + ac + b^2 + bc + c^2 = \tau_2(a, b, c).$$

Therefore, the theorem holds for $m = 4$ and $n = 2$.

VIII-b

We aim to prove the identity:

$$\tau_{m-n}(x_i, \dots, x_{i+n}) = [x_i, \dots, x_{i+n}]x^m,$$

using the recursive relation:

$$\tau_{k+1}(x_1, \dots, x_n, x_{n+1}) = \tau_{k+1}(x_1, \dots, x_n) + x_{n+1}\tau_k(x_1, \dots, x_n, x_{n+1}). \tag{3.54}$$

We proceed by induction on n.

Base Case: $n = 0$

Then the left-hand side is:

$$\tau_m(x_i) = x_i^m,$$

since the complete symmetric polynomial in one variable is just that variable raised to the power m. The right-hand side is:

$$[x_i]x^m = x_i^m.$$

Thus, the identity holds for $n = 0$.

Inductive Step

Assume the identity holds for all indices up to $n-1$, i.e.,

$$\tau_{m-(n-1)}(x_j, \ldots, x_{j+n-1}) = [x_j, \ldots, x_{j+n-1}]x^m$$

for any valid j. Now consider n points: $x_i, \ldots, x_{i+n}$. By the recursive definition of divided differences:

$$[x_i, \ldots, x_{i+n}]x^m = \frac{[x_{i+1}, \ldots, x_{i+n}]x^m - [x_i, \ldots, x_{i+n-1}]x^m}{x_{i+n} - x_i}.$$

By the inductive hypothesis:

$$[x_{i+1}, \ldots, x_{i+n}]x^m = \tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n}),$$

$$[x_i, \ldots, x_{i+n-1}]x^m = \tau_{m-n+1}(x_i, \ldots, x_{i+n-1}).$$

Substituting these into the divided difference expression:

$$[x_i, \ldots, x_{i+n}]x^m = \frac{\tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n}) - \tau_{m-n+1}(x_i, \ldots, x_{i+n-1})}{x_{i+n} - x_i}.$$

Now, we use the recursive relation for complete symmetric polynomials. Consider the identity:

$$(x_{n+1} - x_1)\tau_k(x_1, \ldots, x_{n+1}) = \tau_{k+1}(x_2, \ldots, x_{n+1}) - \tau_{k+1}(x_1, \ldots, x_n),$$

which can be derived from (3.54). Let $k = m - n$, and take the variables to be $x_i, x_{i+1}, \ldots, x_{i+n}$. Then:

$$(x_{i+n} - x_i)\tau_{m-n}(x_i, \ldots, x_{i+n}) = \tau_{m-n+1}(x_{i+1}, \ldots, x_{i+n}) - \tau_{m-n+1}(x_i, \ldots, x_{i+n-1}).$$

Therefore:

$$[x_i, \ldots, x_{i+n}]x^m = \frac{(x_{i+n} - x_i)\tau_{m-n}(x_i, \ldots, x_{i+n})}{x_{i+n} - x_i} = \tau_{m-n}(x_i, \ldots, x_{i+n}).$$

This completes the induction.